



Hybrid modeling for improved extrapolation and transfer learning in the chemical processing industry

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ABSTRACT

In the Chemical Processing Industry, shifts in market demand often require the implementation of new operational modes that balance economic advantages with the need to meet quality and sustainability targets. Hybrid modeling can support these process changes by combining physics-based and data-driven modeling principles to improve prediction accuracy and interpretation. In this work, the use of hybrid modeling for improved extrapolation and transfer learning activities is examined in the context of simulated biodiesel production. We study and compare different configurations of hybrid modeling, including the parallel and serial structures. We also compare hybrid modeling approaches with physics-based and data-driven models, and find that hybrid modeling consistently outperforms these benchmarks in both extrapolation and transfer learning tasks. Hybrid modeling also requires fewer samples than other benchmarks for the transfer learning task. These results suggest that hybrid modeling is an effective approach for supporting decision-making and optimizing process changes in the CPI.

1. Introduction

Chemical Processing Industry (CPI) face a number of challenges in adapting to changing market conditions and demands, when the need to flexibly and rapidly respond to new scenarios is imperative (Kim et al., 2023; Reis and Gins, 2017). Such changing external conditions often result in the implementation of new operational modes that are economically viable while still meeting quality and sustainability targets (Liew et al., 2014). Examples include the need to produce a new product grade or to significantly shift the operation window due to changes in energy, raw material and utility costs (Molinaro and Orzes, 2022; Tomba et al., 2012). In this context, data analytical technologies, such as hybrid modeling, can provide solutions to support and speed up process changes in the CPI. Hybrid modeling has gained significant attention in recent years due to the increasing availability of data and the development of new analytics methodologies. By combining multiple modeling principles and techniques, and taking the most out of all available sources of knowledge in chemical systems (first-principles knowledge, heuristic knowledge, and knowledge derived from process data, (Oliveira, 1998; Schubert et al., 1994)), hybrid modeling frameworks can enable a more com-

prehensive understanding of the process and how it is likely to change over time, enabling better decision-making and proactive management of process operations (de Calle-Etxabe et al., 2020; von Stosch et al., 2014).

Hybrid modeling offers certain advantages over pure physics-based or data-driven approaches. Physics-based models are typically more difficult to develop and require a significant amount of prior knowledge, but are interpretable, often more accurate and have better extrapolation capabilities. Data-driven models, on the other hand, are faster to develop and more flexible, but constrained to the specific data domain in which they are trained and less interpretable (Sansana et al., 2021). The concept of integrating mechanistic and data-driven modeling principles was initially introduced in 1992, whereby equations expressing mass and energy balances were combined with neural networks or radial basis functions (Johansen and Foss, 1992; Kramer et al., 1992; Psychogios and Ungar, 1992; Su et al., 1992). First-principles models and data-driven models can be combined under various hybrid structures, including serial and parallel structures, which will be discussed in greater detail later. The main objective of these architectures is to assemble all available knowledge about a system and create synergies that

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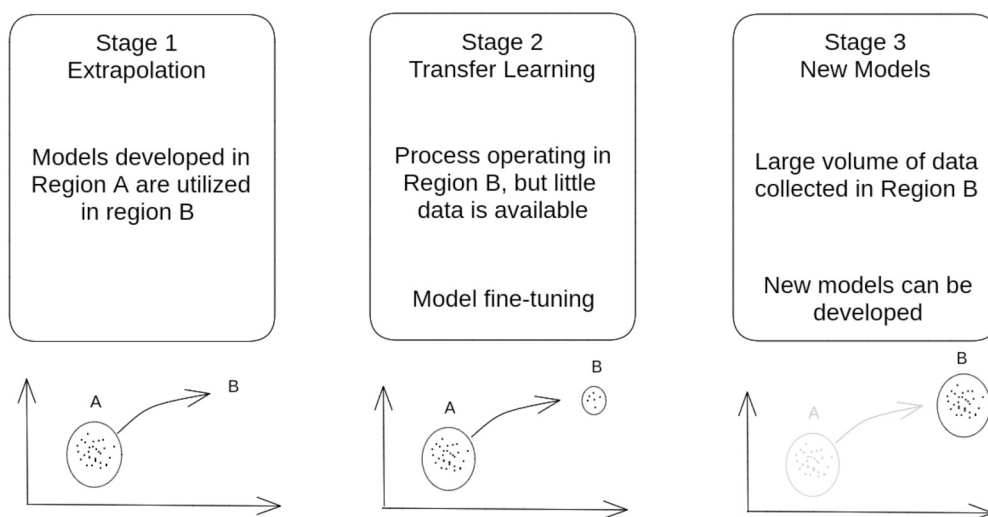


Fig. 1. Implementation of new operating conditions.

lead to improved prediction accuracy and/or stability in the final model (Oliveira, 2004; Zhuang et al., 2022).

Managing process changes in the CPI often involves moving the process from its current operating region (region A) to a new region (region B), as shown in Fig. 1 (Farahani et al., 2021; Quiñones-Grueiro et al., 2019). This process can be conceptually divided into three stages. In the first stage, models developed in region A are directly extrapolated to region B. The accuracy of this extrapolation is important in order to obtain reliable estimates of the quality metric(s) of interest in region B. In the second stage, as the process continues to operate in region B and more data is collected, model retraining can be performed to account for any mismatches between the model and the actual process behavior in region B. Finally, in the third stage, a large volume of data on region B has been collected, allowing for more robust modeling and analysis.

The second stage, where data available in region B are still scarce, is particularly challenging, and methodologies that can efficiently utilize prior knowledge and the limited amount of data are preferred. One approach that has proven useful in these situations is transfer learning (Pan and Yang, 2010), which allows for the leveraging of information obtained in region A to improve the performance of models in region B. Transfer learning can be particularly useful when the underlying process dynamics are similar in both regions A and B, allowing for the transfer of knowledge and experience gained in one region to the other (Facco et al., 2012).

In this work, we investigate the use of hybrid modeling for supporting extrapolation and transfer learning in the context of managing process changes in CPI. We study different configurations of hybrid modeling, including parallel and series configurations, and compare them to benchmarks that include pure physics-based models and data-driven models. The physics-based model used in this study is based on simplified reaction kinetics, while the data-driven methods include partial least squares (PLS) (Wold et al., 2001), least absolute shrinkage and selection operator (LASSO) (Hastie et al., 2016), random forest (RF) (Dietterich, 2000), and neural networks (NN) (Bishop, 2006), to provide reasonable coverage of the data analytics spectrum.

The primary objective of this paper is to demonstrate how hybrid modeling can serve as a valuable tool for industry practitioners to integrate knowledge and data to address evolving process conditions and changing market demands. We argue that the utilization of this technology is still limited due to the scarcity of well-defined guidelines and a perceived steep learning curve for training hybrid models. Throughout the paper, we aim to clarify the advantages of combining “classic” first-principles models with machine learning methods. Furthermore, we elaborate on the application of hybrid modeling in the frame of transfer

learning, illustrating the process of updating an existing hybrid model with new data.

The remainder of the paper is organized as follows: Section 2 introduces several machine learning methods, and revises first-principles modeling and hybrid modeling architectures. In Section 3, a simulated case study from a biodiesel production process is described, as well as the data generation approach followed. This case study was designed to provide a realistic and challenging scenario for testing the proposed hybrid methodologies under new operating conditions. Section 4 presents the results and a discussion of the extrapolation and transfer learning capabilities of the methodologies under analysis. Finally, Section 5 summarizes the conclusions of this work and offers some perspectives for future work.

2. Methods

This section presents the analytical methodologies utilized in this study. Firstly, we discuss the data-driven modeling approaches employed, such as LASSO, PLS, RF, and NN. Next, we present the first-principles model used in this study. Finally, we discuss hybrid model architectures, that combine data-driven and first-principles models.

2.1. Data-driven modeling methodologies

The increasing availability of data in the 21st century, attributed to the continuous development of faster and more informative sensors, larger and distributed storage resources, and greater computational power, has prompted engineers to further consider data as a valuable source of information (Kourti and MacGregor, 1995; Kresta et al., 1991; MacGregor and Kourti, 1995; MacGregor et al., 1994; Reis and Saraiva, 2019). Data science has emerged as a field of knowledge that encompasses statistics, mathematics, and computer science to address the challenges associated with the data-intensive environment usually known as big data. Research on this topic within the context of Process Systems Engineering (PSE) has spurred the development of techniques to address various data-related issues, including high-dimensionality, collinearity, sparsity, nonlinearity, non-stationarity, the presence of noise, outliers, and missing data, among others (Rendall and Reis, 2018). In the following, we describe some of the most commonly used data-driven models within the context of hybrid modeling approaches.

In brief terms, LASSO is a regularization technique that has embedded a variable selection capability by shrinking the less important parameters to zero. PLS is a multivariate regression technique that can handle data with high colinearity by projecting the input dimensions to a low-dimensional predictive subspace of latent variables. Random forest

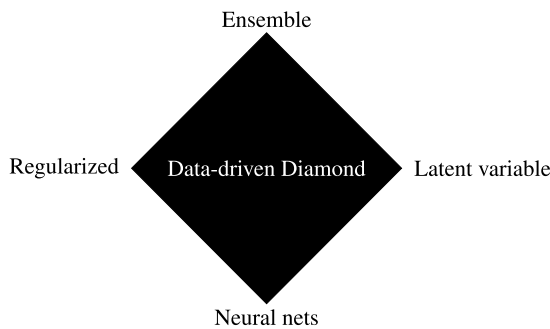


Fig. 2. The “data-driven modeling diamond” considered in this work.

regression is a nonlinear method based on the use of multiple decision trees built up from slightly perturbed subsets of training data, to generate an ensemble predictor of the target response by averaging over all individual predictions. NNs are a type of machine learning model that is inspired by the structure and function of the human brain and is capable of learning complex non-linear patterns from data. These four machine learning methods form what we call the Data-driven Diamond, as shown in Fig. 2, serving as a representation of four types of regression modeling methods: latent variable, ensemble, regularized, and neural networks.

2.1.1. Least absolute shrinkage and selection operator

The LASSO is a regularized modeling technique that is used to estimate sparse coefficients (Hastie et al., 2016). This methodology is particularly useful for datasets with high collinearity and some degree of sparsity regarding variables’ importance for prediction, both of which are frequently encountered in the field of CPI. Due to its ability to generate solutions with fewer non-zero coefficients, the LASSO is also used as a feature selection methodology (Rendall et al., 2019). In general, LASSO provides an estimate for the parameter vector, β , of a multiple regression model represented by the equation:

$$\mathbf{Y} = \mathbf{X} \cdot \beta + \mathbf{E} \quad (1)$$

where $\mathbf{X}$ is the input variables extended matrix (*extended*, because it contains a column of ones to account for the intercept term) or feature matrix, and $\mathbf{E}$ is the residuals matrix. Regularization techniques are widely used in machine learning to stabilize the estimation tasks. By imposing constraints on the magnitudes of the model coefficients, they stabilize the predictions, reducing their variance at the expense of introducing some bias. LASSO is a well-known regularization method that incorporates an L1-norm penalty into the least squares regression problem in order to decrease the magnitudes of the regression coefficients (Tibshirani, 1996). The LASSO solution can be obtained by minimizing the objective function:

$$\hat{\beta}_{\text{LASSO}} = \min_{\hat{\beta}_0, \dots, \hat{\beta}_p} \sum_{i=1}^N (y_i - \hat{y}_i) + \lambda \sum_{j=1}^p |\hat{\beta}_j| \quad (2)$$

where y_i denotes the i^{th} observed response, $\hat{y}_i$ represents the corresponding model prediction, and λ is the hyperparameter that controls the degree of regularization.

2.1.2. Partial least squares

Latent variable methods form a class of modeling approaches that are employed to circumvent challenges associated with high-dimensional, correlated predictors. These techniques operate on the premise that there exists a set of unmeasured variables that drive the variability of the observed ones (Rendall and Reis, 2018). PLS regression establishes a correlation between two sets of data, denoted as $\mathbf{X}$ and $\mathbf{Y}$, by identifying linear combinations of predictors that exhibit maximum covariance with the linear combinations of the responses. The relationship between the two sets can be described by the following equations (Wold et al., 2001):

$$\mathbf{X} = \mathbf{T} \cdot \mathbf{P}^T + \mathbf{E} \quad (3)$$

$$\mathbf{Y} = \mathbf{T} \cdot \mathbf{Q}^T + \mathbf{F} \quad (4)$$

where $\mathbf{T}$ represents the matrix of latent variables, $\mathbf{P}$ represents the X-loadings matrix, $\mathbf{Q}$ represents the Y-loadings matrix, and $\mathbf{E}$ and $\mathbf{F}$ represent the residual matrices for $\mathbf{X}$ and $\mathbf{Y}$ blocks, respectively (Geladi, 1988).

Once the parameters are estimated, the model formed by eqs. (3) and (4) can be recast into a linear regression format, of the type of eq. (1). Although there are various algorithms for estimating a PLS model (Andersson, 2009; de Jong, 1993; Wold, 1975), in this study, we have used the NIPALS algorithm, implemented in scikit-learn (Pedregosa et al., 2011).

2.1.3. Random forest

Regression trees model the relationship between predictors and a response variable by means of a piece-wise constant function. Such a model can predict the value of a target variable by learning simple decision rules from the data features. However, trees tend to have high variance, as small changes in the training set often lead to new trees with different split points and splitting variables. To address these issues, an ensemble of trees is commonly employed to improve prediction performance and decrease prediction variance (Hastie et al., 2016).

RF regression is an algorithm that employs the bootstrap method to generate new datasets, which are used to build multiple regression trees. Prediction for a given sample is obtained by averaging the predictions of all trees in the ensemble (Dietterich, 2000):

$$\hat{y}_i = \frac{1}{N_t} \sum_{n=1}^{N_t} f_n(x_i) \quad (5)$$

where N_t is the number of trees in the ensemble and $f_n(x_i)$ is the prediction from the n^{th} regression tree for the i^{th} observation. Additionally, each tree in RF only considers a randomly selected subset of predictors. The advantage of this method is that the trees in the ensemble are less correlated, leading to increased prediction performance (Rendall et al., 2019).

2.1.4. Neural networks

Neural Networks (NN) are commonly used due to their ability to flexibly model nonlinearities and their independence from prior knowledge about the system being modeled (Oliveira, 2004; Psychogios and Ungar, 1992; Thompson and Kramer, 1994). NNs are composed of a set of interconnected processing units, referred to as neurons, that exchange information between them. There are different types of neural networks, but the most popular for hybrid modeling is the fully-connected feed-forward multilayer perceptron (MLP). Each neuron computes the weighted sum of all incoming signals (S_j) from the previous layer, plus a bias term. The S_j values are called activations and are linear combinations of the input variables (Bishop, 2006). An output is then generated through an activation function, $f(\cdot)$, as follows,

$$\hat{y}_k^{(m)} = f \left(\sum_{i=1}^n \omega_{i,j} X_i + \theta_j \right) \quad (6)$$

where $\omega_{i,j}$ are the weights for neuron i in layer j , X_i are the inputs to neuron i and θ_j is the bias term. There are several activation functions commonly used in neural networks. In this work, we used the rectified linear unit (ReLU) activation function (Maas et al., 2013). For parameter estimation, the Adam optimizer was employed with the goal of minimizing the root mean squared error (MSE) between the training data and model predictions (Kingma and Ba, 2017). The scikit-learn implementation of MLP was adopted (Pedregosa et al., 2011). Although most NNs used in hybrid modeling consist of three layers, it is also possible to employ more layers using deep learning techniques and stochastic gradient descent for the learning process (Schmidhuber, 2015).

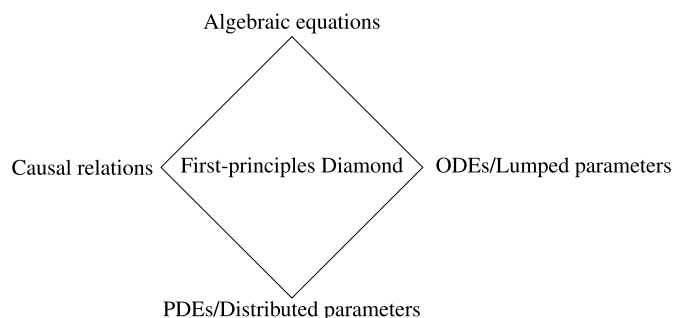


Fig. 3. The “first-principles modeling diamond” considered in this work.

2.2. First-principles modeling methodologies

The construction of a first-principles model for hybrid modeling frameworks depends on the available prior knowledge. This approach involves solving the underlying equations governing the system’s behavior, which may enable a more accurate prediction of the system response. Usually, this knowledge is in the form of balances for extensive system properties, such as mass and energy (Azarpour et al., 2017; Oliveira, 2004; Psychogios and Ungar, 1992) or physical constraints (Gurden et al., 2001; Su et al., 1992; Thompson and Kramer, 1994). These mathematical statements can be expressed more simply as algebraic equations or, with increasing complexity, as ordinary differential equations (ODEs) (for lumped parameter system modeling), differential algebraic equations (DAEs) or PDEs (for distributed parameter system modeling) (von Rueden et al., 2023), as presented in Fig. 3. By increasing the complexity of the mechanistic model, one reduces the structural mismatch between the model and the chemical process, but the larger number of parameters can lead to more uncertain estimates if there is no accurate information for the parameters (this is the bias-variance trade-off for mechanistic models). The first-principles models adopted in this study involve mass and energy balances, and establish a connection between input and output variables. To assess the effectiveness of the hybrid modeling approach, a lumped parameter model was employed as a benchmark or sub-model.

2.3. Hybrid modeling architectures

In the mainstream hybrid modeling literature, the prevalent way to combine the two modeling approaches (first-principles and data-driven) is based on the analysis of the structure of the first-principles model and its assumptions. A parallel cooperative architecture is adopted when there is a need to compensate for a clear mechanistic structural mismatch. On the other hand, if the mechanistic structure is accurate enough, then a serial architecture is usually the preferred choice (von Stosch et al., 2014).

This study aims to compare the performance of the various hybrid model architectures shown in Fig. 4, including the parallel cooperative and serial architectures. Additionally, we include a new fusion-inspired parallel competitive architecture and a reverse serial architecture, which we refer to as *knowledge augmented hybrid model* due to the use of a data augmentation technique. The purpose of this comparison is to determine which architecture is most suitable for a given, a priori defined task, and to identify potential areas for further improvement.

In the CPI, variables are generally continuous, thereby making regression the predominant modeling approach. In the context of hybrid modeling, this assumption remains valid. Our objective is therefore to learn a function $f_H : \mathbb{R}^D \rightarrow \mathbb{R}$ that maps an input with D features, $x \in \mathbb{R}^D$, to a target observation $y \in \mathbb{R}$. To accomplish this, a set of parameters θ , upon which f_H depends, must be learned (the function f_H can be represented as $f_H(x; \theta)$, to make the parameter set explicit).

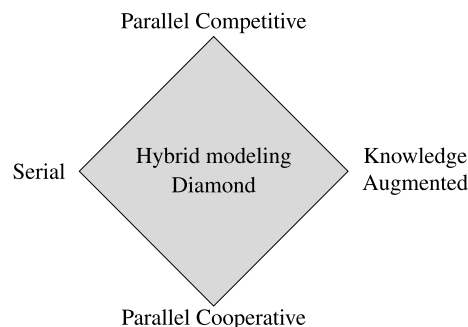


Fig. 4. The “hybrid models diamond” considered in this work.

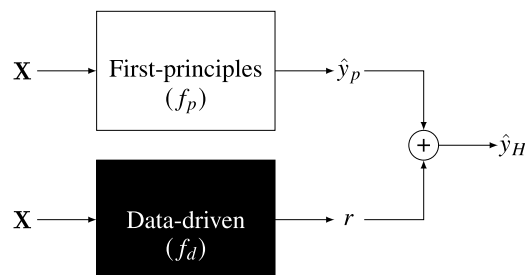


Fig. 5. Schematic representation of the parallel cooperative structure for hybrid modeling.

2.3.1. Parallel cooperative

The parallel cooperative architecture assumes that the function $f_H(x; \theta)$ can be expressed as a summation of a first-principles function $f_p : \mathbb{R}^D \rightarrow \mathbb{R}$ and a data-driven model $f_d : \mathbb{R}^D \rightarrow \mathbb{R}$, $f_p \neq f_d$, as follows:

$$f_H(x; \theta) = f_p(x; \mu) + f_d(x; \psi) \quad (7)$$

where μ represents the parameter set of the first-principles model (f_p) and ψ represents the parameter set of the data-driven model (f_d); θ is the reunion of these two sets, $\theta = \{\mu, \psi\}$. The hybrid model can be trained separately by first identifying μ with respect to the output variable. Subsequently, once f_p is known, the residuals can be computed by using the predictions of the first-principles model, expressed as follows:

$$r_i = y_i - f_p(x; \mu) \quad (8)$$

Finally, the parameters of the data-driven approach ψ can be fitted to the residual dataset $(x_i, r_i)_{i=1}^N$. In summary, the hybrid model operates by improving the first-principles model through the use of a data-driven model that explains part of the structure still present in the residuals, as depicted in Fig. 5. The model predictions are obtained by the following expression:

$$\hat{y}_{H,i} = \hat{y}_{p,i} + r_i \quad (9)$$

where $\hat{y}_{H,i}$ represents the output of the hybrid model, $\hat{y}_{p,i}$ represents the output of the first-principles model, and r_i represents the residuals computed by the data-driven model.

2.3.2. Parallel competitive

In the parallel competitive architecture, the function $f_H(x; \theta)$ can be expressed as a weighted average of a function derived from first-principles $f_p : \mathbb{R}^D \rightarrow \mathbb{R}$, and another obtained by induction from data $f_d : \mathbb{R}^D \rightarrow \mathbb{R}$, $f_p \neq f_d$. The equation describing this architecture can be presented as follows:

$$f_H(x; \theta) = \alpha f_p(x; \mu) + (1 - \alpha) f_d(x; \psi) \quad (10)$$

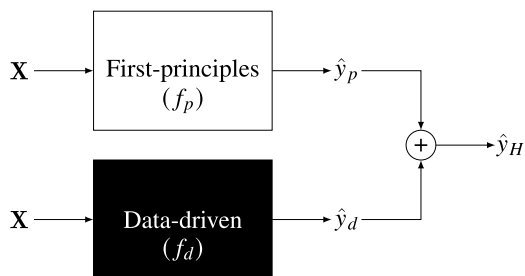


Fig. 6. Schematic representation of the parallel competitive structure for hybrid modeling.

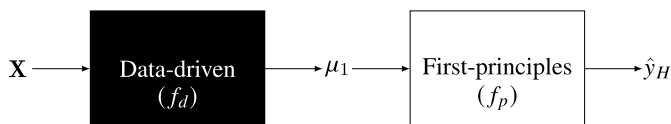


Fig. 7. Schematic representation of the serial structure for hybrid modeling.

The weighting factor, α , which determines the relative importance of each model, is an element of the parameter set θ , i.e., $\theta = \{\alpha, \mu, \psi\}$. In this architecture, both models are trained independently with respect to the target variable y . Subsequently, the weighting factor α is tuned by least-squares minimization. The final model prediction, shown in Fig. 6, has the following form:

$$\hat{y}_{H,i} = \alpha \hat{y}_{p,i} + (1 - \alpha) \hat{y}_{d,i} \quad (11)$$

where $\hat{y}_{H,i}$ represents the output of the hybrid model, $\hat{y}_{p,i}$ represents the output of the first-principles model, and $\hat{y}_{d,i}$ represents the output of the data-driven model.

2.3.3. Serial hybrid

The serial hybrid architecture assumes that the function $f_H(x; \theta)$ is defined by a first-principles model, denoted by $f_p : \mathbb{R}^D \rightarrow \mathbb{R}$, where some or all of its parameters are specified by a data-driven model, represented by $f_d : \mathbb{R}^D \rightarrow \mathbb{R}^P$, where P is the number of parameters to be estimated by f_d and inserted in f_p . The equation governing this architecture can be expressed as follows:

$$f_H(x; \theta) = f_p(x; f_d(x; \psi), \mu), \quad (12)$$

where $f_d(x; \psi)$ denotes the parameters of the first-principles model (f_p) that are learned by f_d , while μ represents the remaining parameters of the first-principles model. The hybrid model can be trained concurrently, with the objective of identifying both μ and ψ with respect to the output variable, denoted by y . This training approach may pose some numerical challenges if the first-principles model f_p is complex.

Alternatively, we will adopt a sequential identification route, which we refer to as the explicit serial hybrid approach. In this training methodology, the first-principles model f_p is solved for $f_d(x; \psi)$ at each available measurements' time, $\{x_i\}_{i=1}^N$, leading to $\{f_d^*(x_i; \psi)\}_{i=1}^N$. Subsequently, the parameters of the data-driven model, ψ , are fitted to this extended dataset represented by $\{x_i, f_d^*(x_i; \psi)\}_{i=1}^N$. Once the data-driven model f_d is determined, it will provide the parameter estimates for the first-principles model, which, in turn, produces the final model prediction, as shown in Fig. 7. The sequential identification route compels the practitioner to inspect the predictions of physics-based parameters from the data-driven model, thereby rendering a more interpretable and transparent approach.

2.3.4. Knowledge augmented hybrid

We propose a new architecture, called knowledge augmented hybrid architecture, which involves augmenting a process dataset with physics-based knowledge. This data augmentation operation is achieved

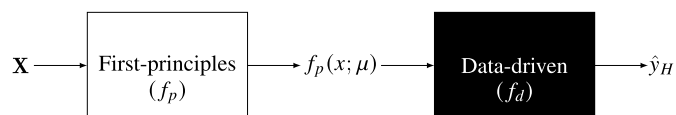


Fig. 8. Schematic representation of the knowledge augmented structure for hybrid modeling.

by concatenating the state predictions obtained from a first-principles model to the existing data matrix. The approach assumes that the function $f_H(x; \theta)$ can be defined by a data-driven model, $f_d : \mathbb{R}^{(D+S)} \rightarrow \mathbb{R}$, where its S inputs are generated by a first-principles model represented by $f_p : \mathbb{R}^D \rightarrow \mathbb{R}^S$. The architecture's governing equations can be expressed as follows:

$$f_H(x; \theta) = f_d(\tilde{x}; \psi), \quad (13)$$

$$\tilde{x} = [x, f_p(x; \mu)], \quad (14)$$

where, $\tilde{x}$ denotes the dataset of process measurements that are augmented with first-principles state predictions. In this architecture, both models are trained independently with respect to the target variable, y .

2.4. Hybrid modeling for transfer learning

In CPI, transfer learning can be effectively employed to adapt or refine models for expanding their use to new product grades or alternative regions of operation. Moreover, transfer learning can facilitate the application of models developed from data collected in pilot units to industrial-scale equipment, or alternatively transfer a model from one equipment, say A, to another similar equipment, B.

The current research investigates the use of hybrid modeling through transfer learning, a methodology we believe has not been extensively documented in existing chemical engineering literature. It is based on the assumption that although the underlying thermodynamics and physical laws governing a chemical process remain unchanged when operating conditions are altered, a model may not be able to fully capture them, being always a simplified version of reality. As George Box famously stated, "All models are wrong...", and therefore, a model that performs well on a particular set of conditions, say A, may not continue to perform well on a different scenario, say B. The discrepancy between the model and the actual process can usually be attributed to one of the following three reasons: 1) parametric mismatch, 2) structural mismatch, or 3) both parametric and structural mismatch (Chen and Ierapetritou, 2020).

The selection of a hybrid structure represents a critical decision when transferring a model, and this choice is strongly influenced by the accuracy of the available model. If the model, regardless of whether it is first-principles, data-driven, or hybrid, exhibits a high degree of structural dissimilarity with the real process, the parallel cooperative architecture (see Fig. 5) is often preferred because the data-driven submodel learns the residuals to correct mismatch. However, the knowledge augmented hybrid architecture (see Fig. 8) is also a viable approach for bridging the discrepancy. In contrast, the serial approach is ill-suited for model transfer in this conditions, as it requires the identification of the subsystem where the knowledge gap exists or of the parameters that require updating (Sansana et al., 2021).

3. Case study

This section presents the case study, beginning with a description of the process simulator used and how data is generated to achieve the objectives of this study. We then present the first-principles model that will play the role of benchmark method, and that will be also used in the construction of the hybrid models.

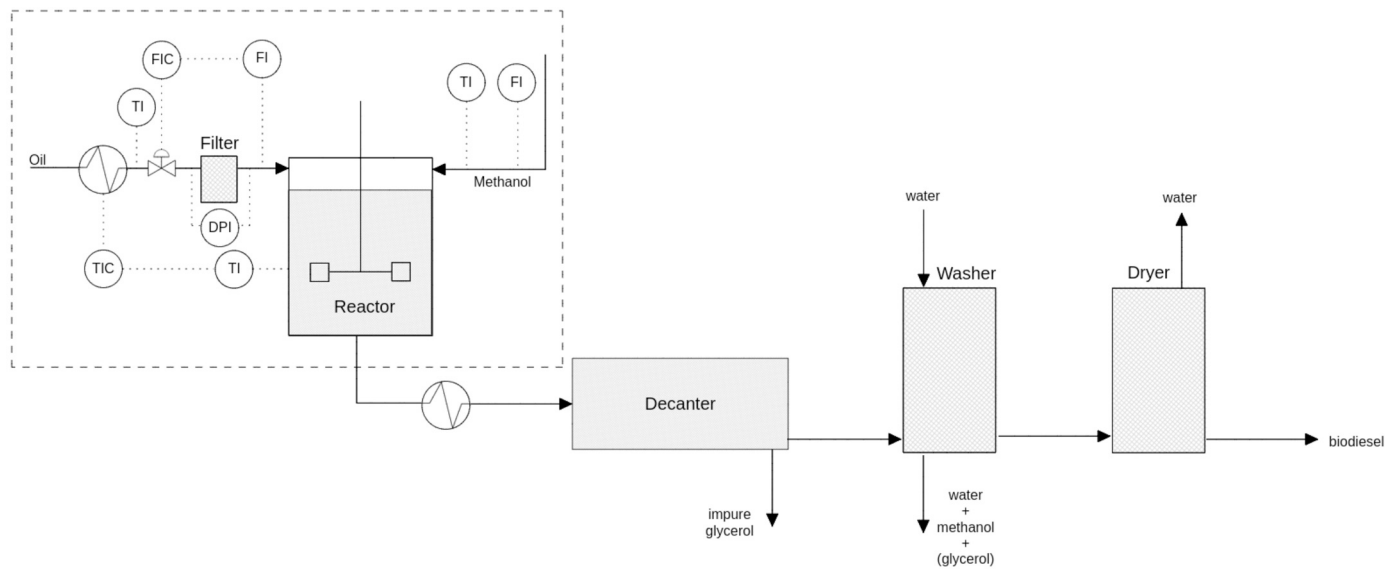


Fig. 9. Process flow diagram of the biodiesel production process simulated in BDsim (adapted from Fernandes et al. (2019)). In the case study, we have only considered the reactor section.

3.1. Data generation

The software BDsim, an open-source simulator of an industrial continuous process for biodiesel production (Brásio et al., 2016; Fernandes et al., 2019), provides the testing scenarios in this study. In particular, we focus on the reaction section of the biodiesel production process, as illustrated in Fig. 9. Data is generated using a 2-level full factorial design with two factors and two center points. The two factors that can be controlled in BDsim's reactor are the reactor temperature (T_R) and the oil feed flow (F_{oil}), which results in 4-month long experimental period for the training set plus a 2-month long experimental period for the center points, totaling 6 months of collected data. The center points represent the simulator's original normal operating conditions, while the upper and lower bounds are obtained by perturbing the center points conditions with $\pm 5\%$ deviations on both input variables, thereby enriching the information content of the dataset.

To evaluate the performance of the models, two test sets were generated: an in-domain test set (ID test) and an out-of-domain test set (OOD test), as depicted in Fig. 10. The former comprises an additional month simulated with input variables at an intermediate level of $+2.5\%$ deviation. The latter, generated from a 2-level full factorial design with two factors, includes only one middle point at the beginning and increased deviations of $\pm 10\%$. We use the OOD test set to assess the extrapolation performance (stage I) and transfer learning capabilities (stage II).

The data points were sampled at an hourly rate, with seven variables being monitored: temperature of the reactor (T_R), mass flow-rate of oil feed (F_{oil}), pressure drop across the oil filter (ΔP), mass flow-rate of methanol feed (F_m), temperature of oil (T_{oil}), temperature of the methanol stream (T_m), and the biodiesel molar fraction in the reactor output stream (x_{biod} , which is also the KPI).

3.2. First-principles model (benchmark)

The behavior of a continuous stirred tank reactor (CSTR) can be modeled using a system of mass balance equations that describe the concentrations of the components of interest in the liquid phase. Under the assumption that the liquid phase is uniformly mixed, the mass balance equations can be formulated in the state space representation as follows:

$$\frac{d\mathbf{x}}{dt} = (N_m(\mathbf{x}_m - \mathbf{x}) + N_o(\mathbf{x}_o - \mathbf{x}) + \mathbf{r}V_R)/n_R, \quad (15)$$

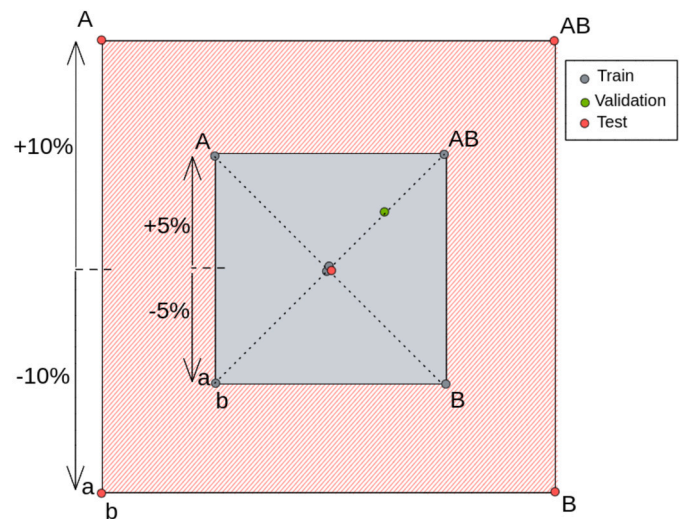


Fig. 10. Representation of the operating conditions simulated for generating datasets.

where $\mathbf{x}$, $\mathbf{x}_m$, and $\mathbf{x}_o$ denote the vectors of molar fractions of component i in the reactor, methanol input stream, and oil input stream, respectively. The input methanol molar flow and oil molar flow are represented by N_m and N_o , respectively. The reaction rate of component i is denoted by the vector $\mathbf{r}$, V_R represents the reactor's volume, and n_R is the total number of moles in the reactor.

The mass balance equations for the system under consideration, where the reaction



takes place, are composed of four equations. The dynamics of the system can be described by the following equation for temperature (energy balance):

$$\frac{dT_R}{dt} = \frac{N_m C_{Pmol,m}(T_m - T_R) + N_o C_{Pmol,o}(T_o - T_R) + r_i V_R}{n_R C_{Pmol,R}}, \quad (17)$$

where T_R , T_m , and T_o denote the temperatures of the reactor, methanol stream, and oil stream, respectively. $C_{Pmol,m}$, $C_{Pmol,o}$, and $C_{Pmol,R}$ represent the molar heat capacities of the methanol stream, the oil stream,

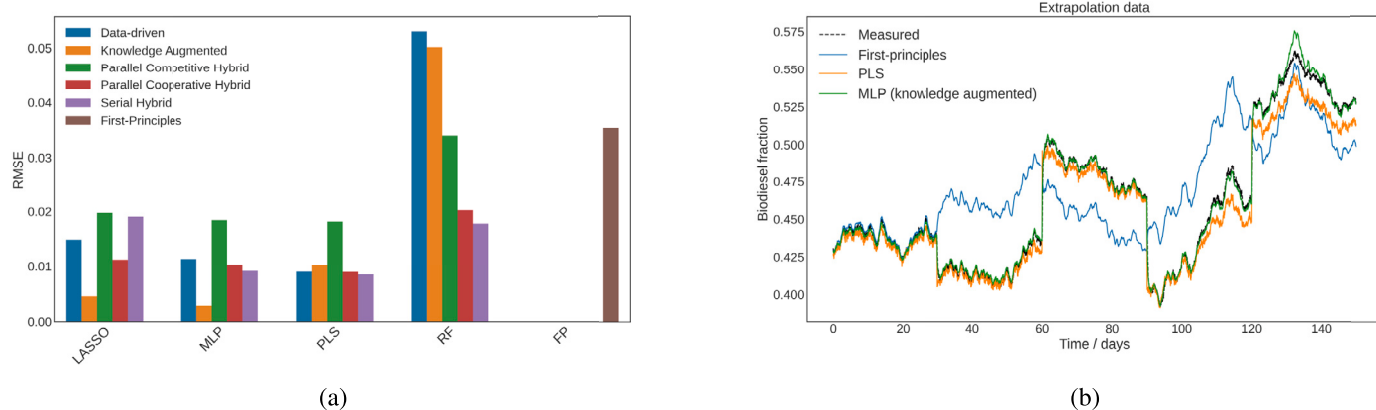


Fig. 11. a) RMSE of all the models in extrapolation; b) Biodiesel fraction and best models predictions in OOD test (extrapolation).

and the reactor, respectively. The reaction rate vector, denoted by $\mathbf{r}$, is defined as

$$\mathbf{r} = [-r, -3r, r, 3r], \quad (18)$$

where r is the reaction rate, a pseudo-parameter to be estimated from data.

4. Results

This section presents the results obtained from the models trained on different application scenarios. The results will be presented in two stages, one regarding an extrapolation scenario (Stage I) and the other a transfer learning scenario (Stage II). More specifically, in Stage I we demonstrate the process of training and validating the models and evaluate their ability to extrapolate to new operating conditions. In Stage II, we utilize one week of process measurements data to perform transfer learning using a hybrid modeling approach based on the best model from Stage I. We then compare this approach with a data-driven model trained using the new week's data.

4.1. Stage I: extrapolation

During Stage I of the modeling process, it is postulated that the availability of data concerning a specific set of operating conditions or product grades enables the development of data-driven models. Additionally, it is assumed that the underlying process is reasonably understood at an engineering level and thus can be expressed in mathematical terms. Conservation laws governing mass and energy balances are typically utilized to generate these models in the CPI domain, provided that sufficient knowledge is available.

The choice of the hybrid architecture is a critical decision to make when developing a hybrid modeling approach. If the first-principles model has a high structural mismatch with the real process, the parallel cooperative and knowledge augmented hybrid architectures are often a good choice. This kind of structure allows engineers to use linear and static mass and energy balances even in nonlinear and dynamic systems, because the data-driven model will, in principle, account for those effects and accommodate them in the hybrid model. The main disadvantage of the parallel cooperative architecture is the lack of extrapolation properties when compared to the serial structure. Therefore, when the goal is to extrapolate to a new set of process conditions, the serial hybrid architecture, as well as the knowledge augmented hybrid architecture are expected to be more accurate and robust.

A concise summary of the extrapolation performance of the models is displayed in Fig. 11a. The most effective model for extrapolation is the knowledge augmented hybrid model utilizing the MLP, which demonstrated only 8.6% of the RMSE presented by the first-principles model

and 33% of the RMSE of the best data-driven model. In Fig. 11, we illustrated the measured KPI, the first-principles model, the best data-driven model, and the most effective hybrid model. The hybrid model tracks the measured KPI almost perfectly, with the exception of the final peak value where it overestimates the measured KPI. Next, we present more details for the more relevant models.

4.1.1. First-principles model

Equations (15)–(18) provide a sound characterization of the process described in Section 3.1 and illustrated in Fig. 9. The measured KPI is the biodiesel fraction in the output stream of the reactor. Assuming an ideal mixing scenario in the CSTR, the diesel fraction is equivalent to x_B in the first-principles model introduced in Section 3.2. Nonetheless, to fully define the model, the reaction rate r must be identified.

Given the abundance of data available and the single degree of freedom for model fitting, parameter identification is a relatively straightforward task. The reaction rate is estimated by minimizing the squared error between the measured biodiesel fraction and the corresponding predicted value using the GEKKO optimization suite (Beal et al., 2018). The identified value of r is $0.0479 \text{ mol} \cdot \text{m}^{-1}$. The resulting model and the training region are depicted in Fig. S.1a, where the model exhibits a root mean squared error (RMSE) of 0.015 and a coefficient of determination (R^2) of 0.44. When it comes to the performance of the first-principles model over unseen conditions, the test RMSE and R^2 are 0.011 and -1.476 , respectively. On extrapolation, the goal of Stage I, $\text{RMSE} = 0.035$ and $R^2 = 0.388$. The results are shown in Figs. S.1b and S.1c. The high RMSE observed in the first-principles model suggests a fundamental discrepancy. This mismatch primarily stems from the simpler kinetic description adopted in the model, as delineated in Section 3.2. In light of these observations, practitioners are usually presented with two options: one avenue is the refinement of the existing first-principles model to improve the level of fundamental mechanistic understanding of the system and its kinetics; alternatively, as explored in our study, one may opt to leverage existing data through hybrid modeling.

4.1.2. Data-driven models

The training of the data-driven models involves the use of six predictor variables: reactor temperature (T_R), oil feed mass flow rate (F_{oil}), oil filter pressure drop (ΔP), methanol feed mass flow rate (F_m), oil temperature (T_{oil}), and methanol stream temperature (T_m). The models were developed to predict the biodiesel molar fraction (x_{bio} or KPI), which served as the designated target variable. The models' hyperparameters were identified using 10-fold cross-validation by minimizing the cross-validated RMSE between the predicted and measured KPI. The scikit-learn Python software library (Pedregosa et al., 2011) was employed for this task.

After the training stage, the models' performance was evaluated on three distinct datasets: the training dataset, the in-domain (ID) test

Table 1
Model performance of every data-driven model in the three datasets.

	R^2				RMSE			
	LASSO	PLS	RFR	MLP	LASSO	PLS	RFR	MLP
Train	0.980	0.986	0.998	0.986	0.003	0.002	0.001	0.002
ID Test	0.922	0.902	0.220	0.907	0.002	0.002	0.006	0.002
OOD Test	0.893	0.960	-0.369	0.937	0.015	0.009	0.053	0.011

Table 2
Model performance of every serial hybrid model in the three datasets.

	R^2				RMSE			
	LASSO	PLS	RFR	MLP	LASSO	PLS	RFR	MLP
Train	0.829	0.997	0.996	0.950	0.008	0.001	0.001	0.005
ID Test	0.050	0.830	0.077	-1.456	0.007	0.003	0.006	0.012
OOD Test	0.822	0.964	0.846	0.959	0.019	0.009	0.018	0.009

dataset, and the out-of-domain (OOD) test dataset (see Fig. S.2). The evaluation of the models' performance on the training dataset assesses their ability to replicate previously observed data (*i.e.*, the fitting ability). In the ID test, we varied process conditions, while ensuring that the KPI remained within the limits of the training dataset. The OOD test, on the other hand, is representative of an extrapolation situation, which is the main focus of this study. The performance results are presented in Table 1. The PLS model was found to be the most effective data-driven model for extrapolation, although LASSO and MLP also exhibited favorable predictive capabilities. PLS' superiority over the other models implies that the association between predictors and target is mostly linear, in addition to the fact that the predictor's variables might exhibit some degree of collinearity. It should be noted that while the random forest regression (RFR) model was the most accurate for the training dataset, it performed poorly under extrapolation.

4.1.3. Explicit serial hybrid models

In the serial hybrid architecture applied to an extrapolation scenario, the data-driven models PLS and MLP have demonstrated superior accuracy. The LASSO method also exhibited acceptable performance. Furthermore, it is noteworthy that the utilization of RFR as the data-driven submodel has yielded favorable performance metrics. This can be attributed to the fact that the predictive capability arises primarily from the first-principles model, and the data-driven component is specifically concerned with predicting the reaction rate as an input to the former model. Thus, the serial hybrid model is very well suited for cases where there is a high level of confidence in the first-principles model, except on a well-defined subsystem that can be supplied through a properly chosen machine learning model. The results are summarized in Fig. S.5, while the performance metrics are provided in Table 2.

It is important to note that identifying a serial hybrid model explicitly confers a significant advantage when compared to the implicit (simultaneous) alternative. By initially computing the reaction rates, the practitioner is encouraged to visualize these reaction rates. This, in turn, enhances the confidence in the serial hybrid model, as it indicates that the data-driven model is not merely predicting a parameter that yields good prediction performance, but rather, a parameter that is consistent with engineering knowledge. Moreover, it is possible to plot the reaction rates alongside the relevant process variables, thereby potentially revealing previously unknown insights into the operation of the process and its effect on critical parameters. In the present case study, the reaction rates were plotted as functions of the methanol feed flow, oil feed flow, reactor temperature, and the interaction between the reactor temperature and oil feed flow (refer to Fig. 12). In Fig. 12a, it can be observed that the methanol feed flow has a positive correlation with the reaction rate for each of the depicted operation levels. Additionally, examining Fig. 12b reveals that these levels are obtained from the oil feed flow, which exhibits a strong positive effect on the reaction rates. As ex-

pected from engineering knowledge, the reactor temperature, which is related to the reaction rate (by the Arrhenius equation), is positively correlated to the reaction rate for the same level of oil flow (see Fig. 12c). Furthermore, the interaction of the two process variables with the reaction rate is also positive, as illustrated in Fig. 12d.

4.1.4. Knowledge augmented hybrid models

As detailed in Section 2.3, knowledge augmented hybrid models comprise the training of a data-driven model on a dataset of process measurements that has been augmented with the states predicted by a first-principles model. This model has been previously trained and applied, enabling access to its predictions that may extend beyond the original operation window. In our specific case study, the first-principles model generates 5 state variables, all located within the reactor: biodiesel molar fraction, methanol molar fraction, glycerin molar fraction, oil molar fraction, and reactor temperature. These 5 state variables are concatenated with the process measurements, and the resulting augmented dataset is used to train the data-driven models. The training process of these models involves the minimization of the cross-validated RMSE between the predicted and computed KPIs (10-fold cross-validation). The outcomes are presented in Fig. S.6, while the metrics that reflect the performance of the model can be found in Table 3.

The knowledge augmented hybrid model featuring MLP as the data-driven model was found to be the best performing model for extrapolation, but LASSO and PLS also presented very good predictive capabilities. In this architecture, RFR showed a bad performance, possibly because it overfitted the training data.

4.2. Stage II: transfer learning

In Stage II, the process is assumed to have moved its operation window from the initial mode, A, to a new set of conditions, B, for a period of time. This period of time is still short and insufficient for generating enough data to establish a new reliable model. For the purposes of this study, a week's worth of data on the new conditions was considered to be available. As explained in Section 3, the OOD test comprises four sets of different conditions, thereby allowing Stage II to be tested on diverse datasets. It should be noted that this type of analysis can only be performed using a process simulator, as in reality, a chemical process would only encounter a single set of new conditions at a time. Fig. 13 displays the new conditions, alongside with the predictions of the first-principles model, the best purely data-driven model (PLS), and the best hybrid model (knowledge augmented hybrid architecture with MLP as the data-driven submodel). Notably, all the models were trained exclusively on condition A, which constitutes the training dataset. As observed, model predictions frequently exhibit bias, which may result from parametric or structural mismatch. This mismatch is more pronounced

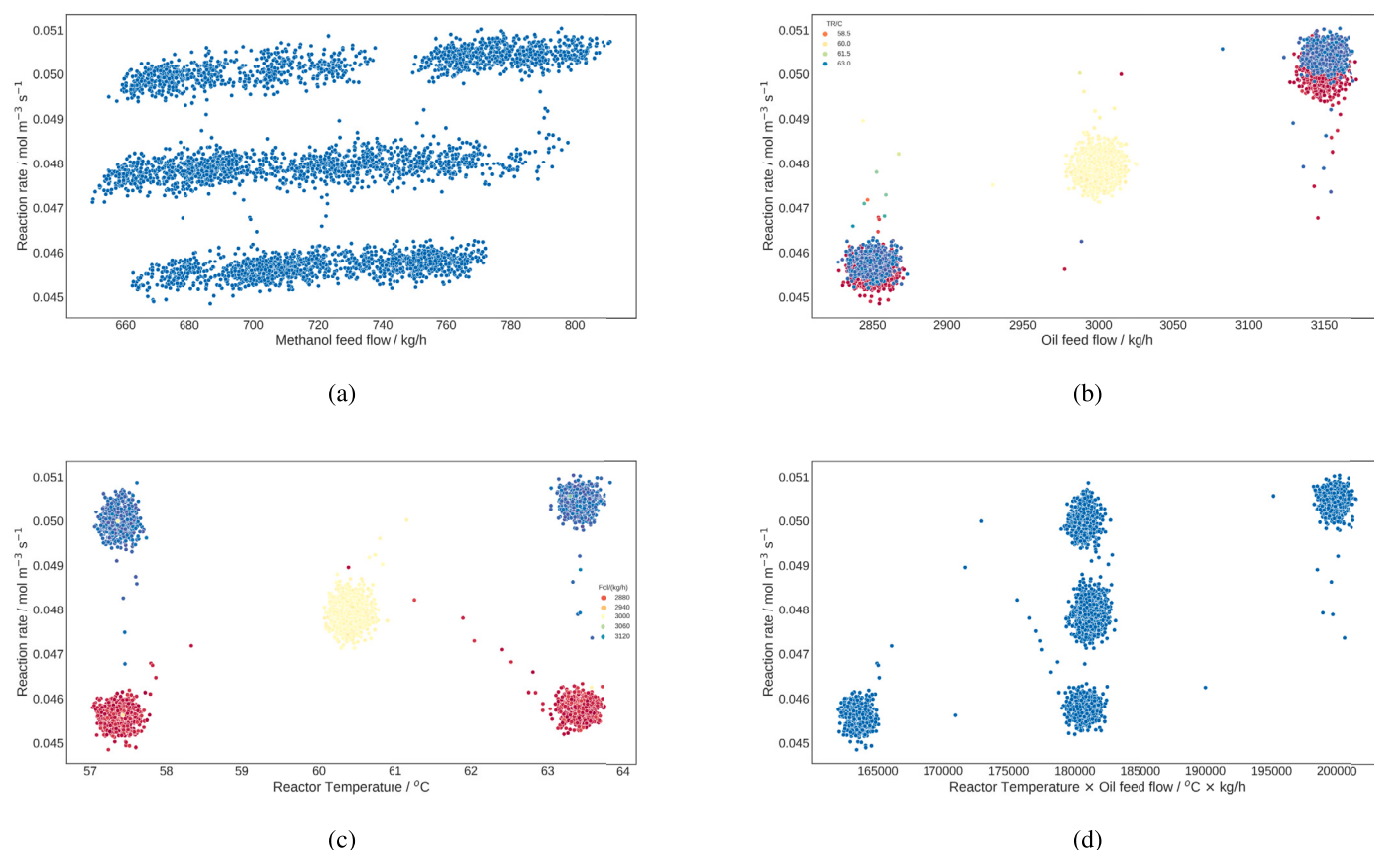


Fig. 12. Instantaneous reaction rates as a function of a) the methanol feed flow; b) the oil feed flow; c) the reactor temperature; d) and the interaction of the reactor temperature with the oil feed flow.

Table 3

Model performance of every knowledge augmented hybrid model in the three datasets.

	R ²				RMSE			
	LASSO	PLS	RFR	MLP	LASSO	PLS	RFR	MLP
Train	0.992	0.997	0.999	0.996	0.002	0.001	0.001	0.001
ID Test	0.829	0.815	0.301	0.868	0.003	0.003	0.006	0.002
OOD Test	0.990	0.950	-0.227	0.995	0.005	0.010	0.050	0.003

in the first-principles model, whereas the hybrid model displays less off-set.

In order to mitigate model mismatch, we have employed a hybridization approach, where we utilize the previously trained hybrid model (referred to as the source model) to generate a target model. As described in Section 2.4, the most suitable architectures for this purpose are the knowledge augmented hybrid (TL-KAH) and the parallel cooperative (TL-PH). In the former architecture, the prediction from the source model is incorporated as a variable into the process measurements available. This augmented dataset is then used to correct the source prediction, thereby aligning it with the target conditions. To make the prediction, we employ the PLS model, as it was found to be the best data-driven model in several occasions in the previous subsection (Stage I). For the TL-PH architecture, we build a PLS model to predict the error between the source model and the target measurement, thus correcting the initial predictions. As benchmarks for these models, we compare them against the source model without any changes and a PLS model trained using the newly available week of data. The resulting RMSEs are presented in Fig. 14a and the profiles are depicted in Fig. 14b.

The results obtained through the application of hybrid modeling for model transfer were found to be encouraging. The transfer learning models outperformed the source model (KAH_MLP) in Test scenarios 1, 3, and 4. However, in Test scenario 2, the source model was found

to be slightly better. The reason for this can be attributed to the fact that the source model already had a low RMSE and the addition of model complexity via model transfer only led to an increase in variance, thereby degrading its performance. When comparing the performance of the hybrid models with that of PLS, it was observed that in the worst-case scenario (Test 4), TL-PH matched the performance of PLS. In the other scenarios, model transfer yielded significantly better results, thus demonstrating that incorporating available information from previous process conditions added value to the model. Additionally, in three out of four scenarios, the knowledge augmented hybrid architecture outperformed the parallel cooperative architecture. In summary, the use of hybrid modeling techniques for transfer learning was found to be effective, provided that the source model actually needs to be transferred. If the source model extrapolates well, it may be advisable to leave it as it is and monitor its performance.

5. Conclusions

We demonstrated the effectiveness of hybrid modeling as a tool for industry practitioners to address evolving process conditions and market demands by integrating knowledge and data. While the utilization of this technology is still limited due to the scarcity of well-defined instructions and the perceived steep learning curve for training hybrid

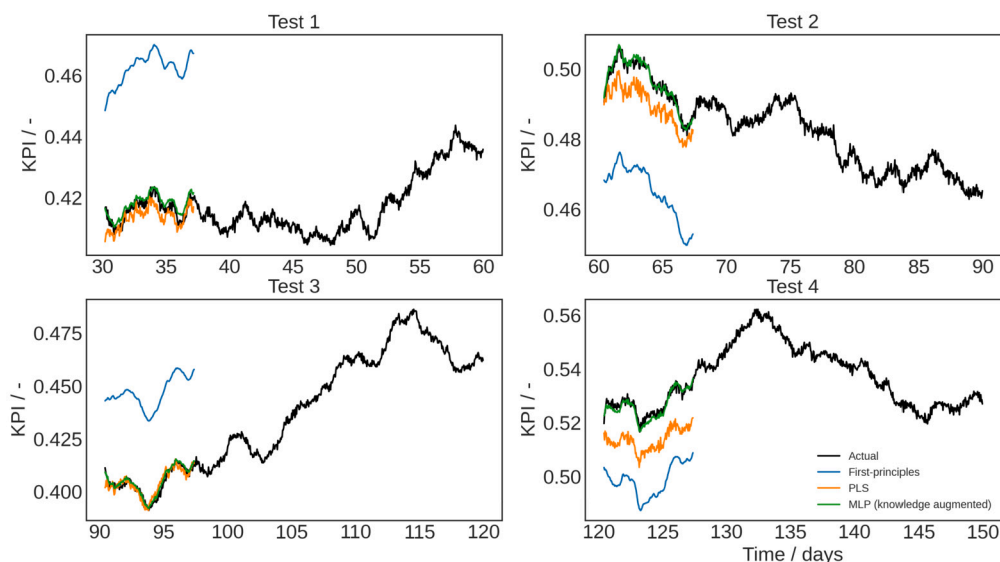


Fig. 13. Implementation of new operating conditions.

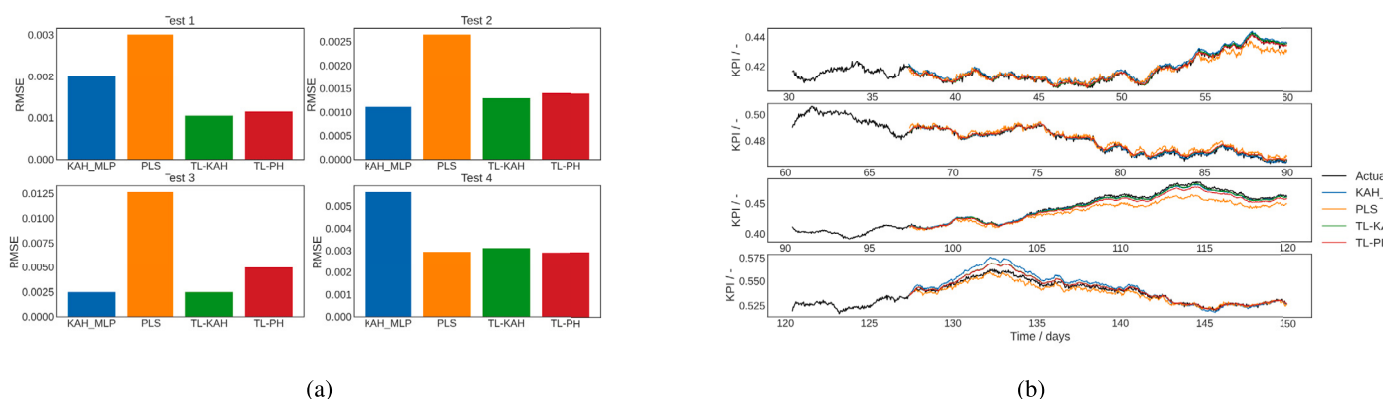


Fig. 14. a) RMSE of transferred models and benchmarks; b) Biodiesel fraction and models predictions.

models, we believe the advantages of combining first-principles models with machine learning methods are solid and go beyond the increased interpretability when compared to data-driven models, as demonstrated in this article.

The hybrid modeling methodology was applied to a simulated case study of biodiesel production operated under different conditions that induce changes in biodiesel concentration (Fernandes et al., 2019). The performance of different modeling approaches, including hybrid modeling, pure physics-based models, and pure data-driven models, was assessed in terms of their ability to accurately predict process behavior and support extrapolation and transfer learning. The results showed that hybrid modeling consistently outperformed the other approaches, with serial hybrid approaches being preferred for the extrapolation task. Hybrid modeling also required fewer samples than the other benchmarks for the transfer learning task. These findings suggest that hybrid modeling is a useful approach for supporting extrapolation and transfer learning in the context of managing process changes in the Chemical Processing Industries.

The choice of the hybrid architecture is a critical decision in developing a hybrid modeling approach. When the first-principles model has a high structural mismatch with the real process, parallel cooperative and knowledge augmented hybrid architectures are often the best choices. This structure enables engineers to use linear and static mass and energy balances even in nonlinear and dynamic systems, with the data-driven model accounting for those effects and accommodating them in the hybrid model. The disadvantage of the parallel cooperative

architecture is its lack of extrapolation properties compared to the serial structure. Thus, the serial hybrid architecture, as well as the knowledge augmented hybrid architecture, are preferred when extrapolation to a new set of process conditions is the goal. In the case study explored in this article, the most effective model for extrapolation is the knowledge augmented hybrid model utilizing the MLP, which demonstrated only 8.6% of the RMSE presented by the first-principles model and 33% of the RMSE of the best data-driven model. The hybrid model accurately tracks the measured KPI, with the exception of the final peak value where it overestimates.

Moreover, the research details the application of transfer learning in conjunction with hybrid modeling, showcasing the adaptation of an existing model to novel process conditions through the training of an additional hybrid layer with a minimal dataset. The use of hybrid modeling techniques for transfer learning was found to be effective, provided that the source model actually needs to be transferred. Transfer learning is recommended when model performance in the new region experiences a significant decline in accuracy.

In the future, we envision developing a systematic framework for developing hybrid models under the principles of Information Quality (InfoQ) (Reis and Kenett, 2018). The aim is to link the quality of the available data, goal definition, and engineering knowledge about a process, to produce the hybrid architecture that generates the highest level of information quality.

CRedit authorship contribution statement

Joel Sansana: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ricardo Rendall:** Writing – review & editing, Validation, Conceptualization. **Ivan Castillo:** Writing – review & editing, Validation, Resources. **Leo Chiang:** Writing – review & editing, Resources. **Marco S. Reis:** Writing – review & editing, Supervision, Resources, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary material

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.ces.2024.120568>.

References

- Andersson, M., 2009. A comparison of nine PLS1 algorithms. *J. Chemom.* 23 (10), 518–529. <https://doi.org/10.1002/cem.1248>.
- Azarpour, A., Borhani, T.N.G., Wan Alwi, S.R., Manan, Z.A., Abdul Mutalib, M.I., 2017. A generic hybrid model development for process analysis of industrial fixed-bed catalytic reactors. *Chem. Eng. Res. Des.* 117, 149–167. <https://doi.org/10.1016/j.cherd.2016.10.024>.
- Beal, L.D.R., Hill, D.C., Martin, R.A., Hedengren, J.D., 2018. GEKKO optimization suite. *Processes* 6 (8), 106. <https://doi.org/10.3390/pr6080106>.
- Bishop, C., 2006. *Pattern Recognition and Machine Learning*. Information Science and Statistics. Springer-Verlag, New York.
- Brásio, A.S.R., Romanenko, A., Fernandes, N.C.P., Santos, L.O., 2016. First principle modeling and predictive control of a continuous biodiesel plant. *J. Process Control* 47, 11–21. <https://doi.org/10.1016/j.jprocont.2016.09.003>.
- Chen, Y., Ierapetritou, M., 2020. A framework of hybrid model development with identification of plant-model mismatch. *AIChE J.*, e16996. <https://doi.org/10.1002/aic.16996>.
- de Calle-Etxabe, K.L., Ruiz-Carcel, C., Starr, A., Ferreira, S., Arnaiz, A., Gomez-Omella, M., Sierra, B., 2020. Hybrid modelling for linear actuator diagnosis in absence of faulty data records. *Comput. Ind. 123*, 103339.
- de Jong, S., 1993. SIMPLS: an alternative approach to partial least squares regression. *Chemom. Intell. Lab. Syst.* 18 (3), 251–263. [https://doi.org/10.1016/0169-7439\(93\)85002-x](https://doi.org/10.1016/0169-7439(93)85002-x).
- Dietterich, T.G., 2000. Ensemble methods in machine learning. In: *Multiple Classifier Systems*. In: *Lecture Notes in Computer Science*. Springer, Berlin, Heidelberg, pp. 1–15.
- Facco, P., Tomba, E., Bezzi, F., García-Muñoz, S., Barolo, M., 2012. Transfer of process monitoring models between different plants using latent variable techniques. *Ind. Eng. Chem. Res.* 51 (21), 7327–7339. <https://doi.org/10.1021/ie202974u>.
- Farahani, H.S., Fatehi, A., Nadali, A., Shoorehdeli, M.A., 2021. Domain adversarial neural network regression to design transferable soft sensor in a power plant. *Comput. Ind. 132*, 103489.
- Fernandes, N., Romanenko, A., Reis, M.S., 2019. Mechanistic modeling and simulation for process data generation. *Ind. Eng. Chem. Res.* <https://doi.org/10.1021/acs.iecr.9b01752>.
- Geladi, P., 1988. Notes on the history and nature of partial least squares (PLS) modelling. *J. Chemom.* 2 (4), 231–246. <https://doi.org/10.1002/cem.1180020403>.
- Gurden, S.P., Westerhuis, J.A., Bijlsma, S., Smilde, A.K., 2001. Modelling of spectroscopic batch process data using grey models to incorporate external information. *J. Chemom.* 15 (2), 101–121. [https://doi.org/10.1002/1099-128x\(200102\)15:2<101::aid-cem602>3.0.co;2-v](https://doi.org/10.1002/1099-128x(200102)15:2<101::aid-cem602>3.0.co;2-v).
- Hastie, T., Tibshirani, R., Friedman, J., 2016. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, 2nd edition. Springer, New York, NY.
- Johansen, T.A., Foss, B.A., 1992. Representing and learning unmodeled dynamics with neural network memories. In: 1992 American Control Conference, pp. 3037–3043.
- Kim, M., Lim, C., Hsuan, J., 2023. From technology enablers to circular economy: data-driven understanding of the overview of servitization and product-service systems in industry 4.0. *Comput. Ind.* 148, 103908.
- Kingma, D.P., Ba, J., 2017. Adam: a method for stochastic optimization. *arXiv:1412.6980*. <https://doi.org/10.48550/arXiv.1412.6980>, Jan. 2017.
- Kourti, T., MacGregor, J., 1995. Process analysis, monitoring and diagnosis, using multivariate projection methods. *Chemom. Intell. Lab. Syst.* 28, 3–21. [https://doi.org/10.1016/0169-7439\(95\)80036-9](https://doi.org/10.1016/0169-7439(95)80036-9).
- Kramer, M.A., Thompson, M.L., Bhagat, P.M., 1992. Embedding theoretical models in neural networks. In: 1992 American Control Conference, pp. 475–479.
- Kresta, J.V., Macgregor, J.F., Marlin, T.E., 1991. Multivariate statistical monitoring of process operating performance. *Can. J. Chem. Eng.* 69, 35–47. <https://doi.org/10.1258/phleb.2011.010101>.
- Liew, W.T., Adhitya, A., Srinivasan, R., 2014. Sustainability trends in the process industries: a text mining-based analysis. *Comput. Ind.* 65 (3), 393–400. <https://doi.org/10.1016/j.compind.2014.01.004>.
- Maas, A.L., Hannun, A.Y., Ng, A.Y., 2013. Rectifier nonlinearities improve neural network acoustic models. In: *ICML*, vol. 30, p. 3.
- MacGregor, J.F., Kourti, T., 1995. Statistical process control of multivariate processes. *Control Eng. Pract.* 3 (3), 403–414. [https://doi.org/10.1016/0967-0661\(95\)00014-1](https://doi.org/10.1016/0967-0661(95)00014-1).
- MacGregor, J.F., Jaeckle, C., Kiparissides, C., Koutoudi, M., 1994. Process monitoring and diagnosis by multiblock PLS methods. *AIChE J.* 40 (5), 826–838. <https://doi.org/10.1002/aic.690400509>.
- Molinaro, M., Orzes, G., 2022. From forest to finished products: the contribution of industry 4.0 technologies to the wood sector. *Comput. Ind.* 138, 103637.
- Oliveira, R., 1998. Supervision, control and optimization of biotechnological processes based on hybrid models. Ph.D. thesis, Ph.D. Dissertation. Martin-Luther Universität, Halle-Wittenberg, Germany.
- Oliveira, R., 2004. Combining first principles modelling and artificial neural networks: a general framework. *Comput. Chem. Eng.* 28 (5), 755–766. <https://doi.org/10.1016/j.compchemeng.2004.02.014>.
- Pan, S.J., Yang, Q., 2010. A survey on transfer learning. *IEEE Trans. Knowl. Data Eng.* 22 (10), 1345–1359. <https://doi.org/10.1109/tkde.2009.191>.
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., Duchesnay, E., 2011. Scikit-learn: machine learning in Python. *J. Mach. Learn. Res.* 12, 2825–2830.
- Psichogios, D.C., Ungar, L.H., 1992. A hybrid neural network-first principles approach to process modeling. *AIChE J.* 38 (10), 1499–1511. <https://doi.org/10.1002/aic.690381003>.
- Quiñones-Grueiro, M., Prieto-Moreno, A., Verde, C., Llanes-Santiago, O., 2019. Data-driven monitoring of multimode continuous processes: a review. *Chemom. Intell. Lab. Syst.* 189, 56–71. <https://doi.org/10.1016/j.chemolab.2019.03.012>.
- Reis, M.S., Gins, G., 2017. Industrial process monitoring in the big data/industry 4.0 Era: from detection, to diagnosis, to prognosis. *Processes* 5 (3), 35. <https://doi.org/10.3390/pr5030035>.
- Reis, M.S., Kenett, R., 2018. Assessing the value of information of data-centric activities in the chemical processing Industry 4.0. *AIChE J.* 64 (11), 3868–3881. <https://doi.org/10.1002/aic.16203>.
- Reis, M.S., Saraiva, P.M., 2019. Data-centric process systems engineering for the chemical Industry 4.0. In: *Systems Engineering in the Fourth Industrial Revolution*. John Wiley & Sons, Ltd., pp. 137–159. Ch. 6.
- Rendall, R., Reis, M.S., 2018. Which regression method to use? Making informed decisions in “data-rich/knowledge poor” scenarios – the predictive analytics comparison framework (PAC). *Chemom. Intell. Lab. Syst.* 181, 52–63. <https://doi.org/10.1016/j.chemolab.2018.08.004>.
- Rendall, R., Castillo, I., Schmidt, A., Chin, S.-T., Chiang, L.H., Reis, M., 2019. Wide spectrum feature selection (WiSe) for regression model building. *Comput. Chem. Eng.* 121, 99–110. <https://doi.org/10.1016/j.compchemeng.2018.10.005>.
- Sansana, J., Joswiak, M.N., Castillo, I., Wang, Z., Rendall, R., Chiang, L.H., Reis, M.S., 2021. Recent trends on hybrid modeling for Industry 4.0. *Comput. Chem. Eng.* 151, 107365. <https://doi.org/10.1016/j.compchemeng.2021.107365>.
- Schmidhuber, J., 2015. Deep learning in neural networks: an overview. *Neural Netw.* 61, 85–117. <https://doi.org/10.1016/j.neunet.2014.09.003>.
- Schubert, J., Simutis, R., Dors, M., Havlík, I., Lübbert, A., 1994. Hybrid modelling of yeast production processes – combination of a priori knowledge on different levels of sophistication. *Chem. Eng. Technol.* 17 (1), 10–20. <https://doi.org/10.1002/ceat.270170103>.
- Su, H.-T., Bhat, N., Minderman, P., McAvoy, T., 1992. Integrating neural networks with first principles models for dynamic modeling. In: *Dynamics and Control of Chemical Reactors, Distillation Columns and Batch Processes*. Elsevier, pp. 327–332.
- Thompson, M.L., Kramer, M.A., 1994. Modeling chemical processes using prior knowledge and neural networks. *AIChE J.* 40 (8), 1328–1340. <https://doi.org/10.1002/aic.690400806>.
- Tibshirani, R., 1996. Regression shrinkage and selection via the Lasso. *J. R. Stat. Soc. B* 58 (1), 267–288. <https://doi.org/10.1111/j.2517-6161.1996.tb02080.x>.
- Tomba, E., Barolo, M., García-Muñoz, S., 2012. General framework for latent variable model inversion for the design and manufacturing of new products. *Ind. Eng. Chem. Res.* 51 (39), 12886–12900. <https://doi.org/10.1021/ie301214c>.

- von Rueden, L., Mayer, S., Beckh, K., Georgiev, B., Giesselbach, S., Heese, R., Kirsch, B., Pfrommer, J., Pick, A., Ramamurthy, R., Walczak, M., Garcke, J., Bauckhage, C., Schuecker, J., 2023. Informed machine learning – a taxonomy and survey of integrating prior knowledge into learning systems. *IEEE Trans. Knowl. Data Eng.* 35 (1), 614–633. <https://doi.org/10.1109/TKDE.2021.3079836>.
- von Stosch, M., Oliveira, R., Peres, J., de Azevedo, S., Feyo, 2014. Hybrid semi-parametric modeling in process systems engineering: past, present and future. *Comput. Chem. Eng.* 60, 86–101. <https://doi.org/10.1016/j.compchemeng.2013.08.008>.
- Wold, H., 1975. Soft modelling by latent variables: the non-linear iterative partial least squares (NIPALS) approach. *J. Appl. Probab.* 12 (S1), 117–142. <https://doi.org/10.1017/s0021900200047604>.
- Wold, S., Sjöström, M., Eriksson, L., 2001. PLS-regression: a basic tool of chemometrics. *Chemom. Intell. Lab. Syst.* 58 (2), 109–130. [https://doi.org/10.1016/s0169-7439\(01\)00155-1](https://doi.org/10.1016/s0169-7439(01)00155-1).
- Zhuang, Y., Liu, Y., Ahmed, A., Zhong, Z., del Rio Chanona, E.A., Hale, C.P., Mercangöz, M., 2022. A hybrid data-driven and mechanistic model soft sensor for estimating CO₂ concentrations for a carbon capture pilot plant. *Comput. Ind.* 143, 103747.